

Customer Churn Prediction – AlphaCom

Final Capstone Project

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Table of Contents

Understanding the Business Problem

- 1.1 Problem Statement
- 1.2 Need of the Study
- 1.3 Business and Industry Context
- 1.4 Business / Social Opportunity

1. Exploratory Data Analysis (EDA)

- 2.1 Data Collection and Background
- 2.2 Data Overview and Description
- 2.3 Univariate Analysis
- 2.4 Bivariate Analysis
- 2.5 Multivariate Analysis
- 2.6 Key Insights from EDA

2. Data Preprocessing

- 3.1 Methodology
- 3.2 Duplicate and Data Cleaning
- 3.3 Anomalous Value Treatment
- 3.4 Missing Value Treatment
- 3.5 Outlier Handling
- 3.6 Feature Engineering
- 3.7 Class Imbalance Handling
- 3.8 Feature Scaling
- 3.9 Data Leakage Prevention

3. Model Building – Baseline Model

- 4.1 Algorithm Selection
- 4.2 Metric of Choice and Rationale
- 4.3 Model Performance
- 4.4 Model Interpretation

4. Model Building – Advanced Models

5. Model Performance Improvement (Hyperparameter Tuning)

6. Model Comparison and Final Model Selection

7. Actionable Insights and Business Recommendations

8. Conclusion

9. Appendix

1. Understanding the Business Problem

1.1 Problem Statement

AlphaCom, a leading telecommunications service provider, has been facing a **persistent increase in customer churn** despite offering a wide portfolio of competitive services. Customer churn refers to customers discontinuing their subscription or switching to a competitor. In a highly saturated and competitive telecom market, churn directly impacts **revenue stability, profitability, and brand reputation**.

Customer churn is influenced by a complex combination of factors such as pricing, contract structure, service quality, billing preferences, and customer demographics. Traditional retention strategies are largely reactive, meaning AlphaCom often responds only after customers leave. This reactive approach results in lost revenue opportunities and higher customer acquisition costs.

The objective of this project is to develop a **data-driven churn prediction framework** that identifies customers at high risk of churn and uncovers the key drivers behind their decisions, enabling AlphaCom to proactively intervene.

1.2 Need of the Study

From a business perspective, **customer retention is significantly more cost-effective than customer acquisition**. Industry studies indicate:

- Acquiring a new customer costs **5–6 times more** than retaining an existing one
- Increasing customer retention by **5% can increase profits by 25%–95%**
- Existing customers have a **60–70% probability** of purchasing new services, compared to **5–20%** for new customers

High churn leads to:

- Loss of recurring monthly revenue
- Increased marketing and onboarding costs
- Reduced customer lifetime value
- Damage to brand credibility

This study is essential to help AlphaCom **shift from a reactive churn management strategy to a proactive, predictive retention strategy**, thereby protecting long-term profitability.

1.3 Business/Social Opportunity

By using analytics and machine learning, AlphaCom can convert churn prediction into a strategic advantage. Accurate churn prediction allows the company to:

- Launch targeted retention campaigns
- Offer personalized discounts and service bundles
- Improve customer experience for high-risk segments
- Increase customer lifetime value
- Reduce overall revenue loss due to churn

The telecommunications industry globally experiences churn rates ranging between **20% and 30% annually**, driven by aggressive competition, low switching costs, and price sensitivity. With multiple service providers offering similar pricing and service bundles, customers can easily migrate if expectations are not met.

In this context, churn prediction is not merely an analytical exercise but a **strategic business necessity**. Companies that leverage predictive analytics can identify vulnerable customers early and deploy targeted retention actions before churn occurs.

2. Exploratory Data Analysis (EDA)

2.1 Data Collection and Background

The dataset contains customer-level information from AlphaCom. Each row represents a unique customer and each column represents a specific attribute related to demographics, service usage, billing preferences, contract type, and churn status. The target variable is **Churn**, indicating whether a customer has left the company.

Variables fall into four main categories:

- Demographics: Gender, SeniorCitizen, Partner, Dependents
- Services: PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
- Billing & Contracts: Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
- Target: Churn

2.2 Data Overview

The initial dataset contained 12,055 rows and 20 columns. After cleaning and removing anomalous records, the final dataset used for analysis contained 11,199 rows.

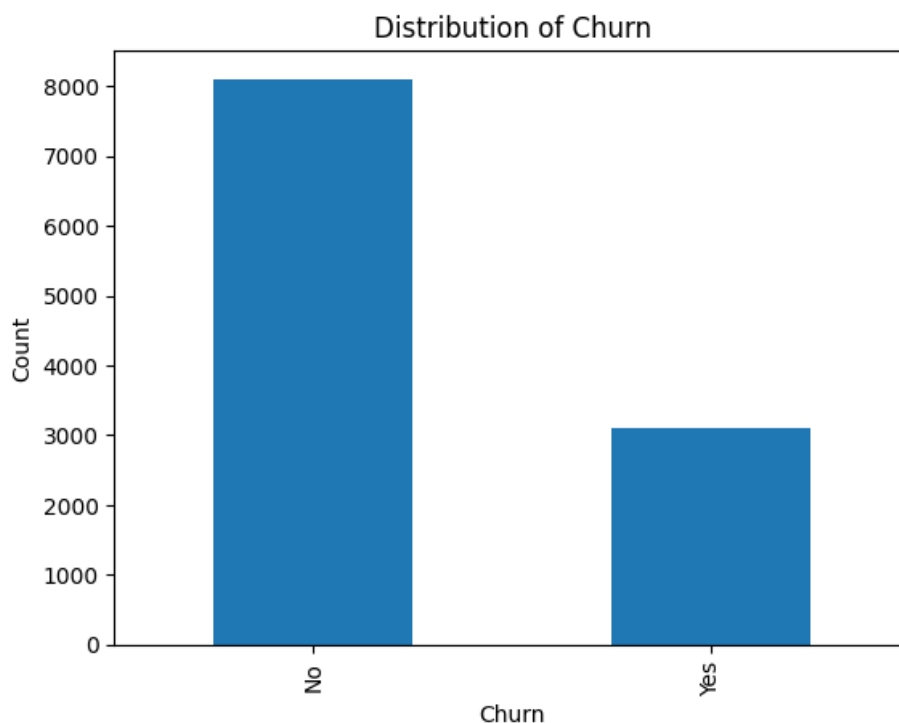
Key observations:

- Target variable has two categories: Yes and No
- About 28% customers have churned and 72% are active
- Dataset includes both categorical and numerical variables
- Monetary columns required cleaning due to currency symbols and formatting issues.

2.3 Univariate Analysis

Univariate analysis examines each variable individually.

- **Churn:** Approximately 28% customers have churned, indicating moderate class imbalance.



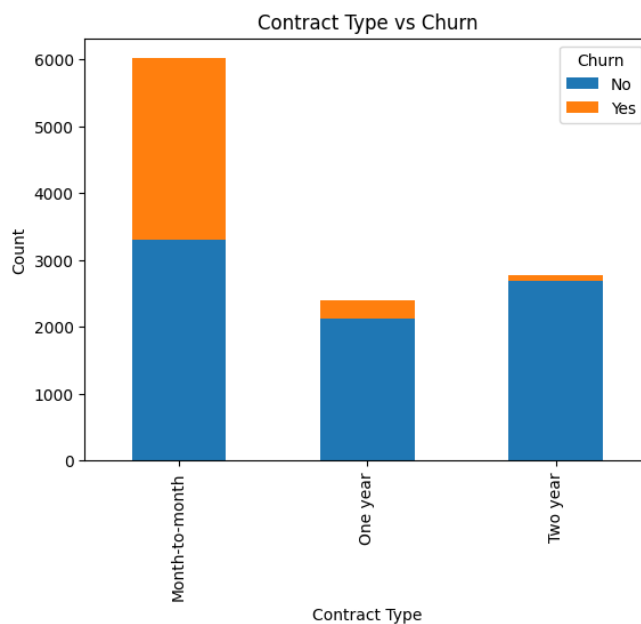
- **Contract:** Most customers are on month-to-month contracts, followed by two-year and one-year contracts.
- **Internet Service:** Fiber optic is most common, followed by DSL and no internet service.
- **Payment Method:** Electronic check is most used, followed by automatic credit card and bank transfer.
- **Numeric Variables:**
 - Tenure ranges from 0 to 74 months
 - MonthlyCharges range from ~\$15 to ~\$122
 - TotalCharges range from ~\$0 to over \$9,000

2.4 Bivariate Analysis

Contract vs Churn

Month-to-month customers show nearly 45% churn, while one-year and two-year contracts show very low churn.

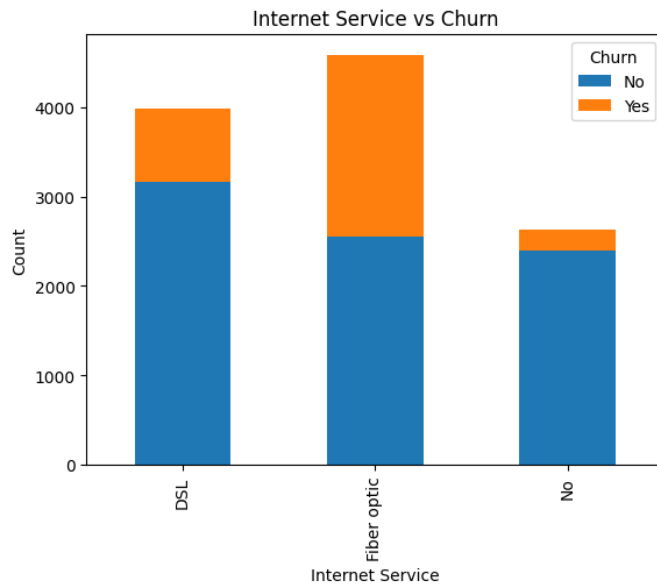
Insight: Short-term contracts strongly increase churn risk.



Internet Service vs Churn

Fiber optic customers have around 44% churn, DSL about 20%, and customers without internet only about 9%.

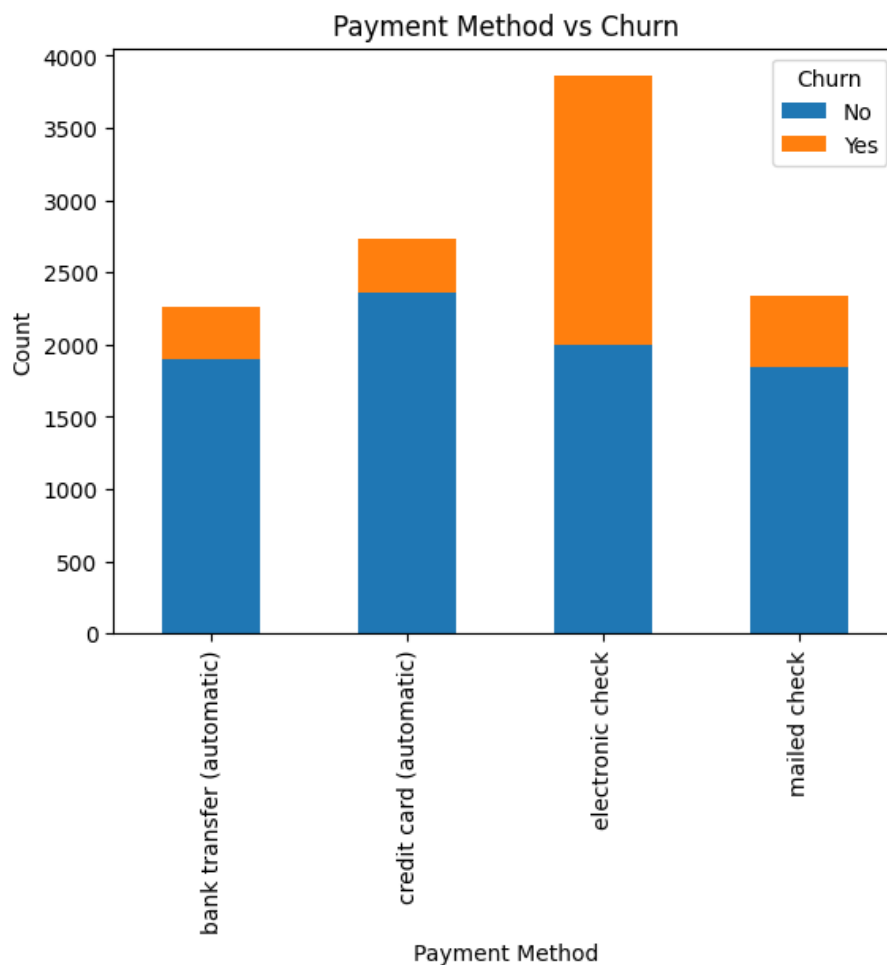
Insight: Fiber optic service likely has pricing or quality issues.



Payment Method vs Churn

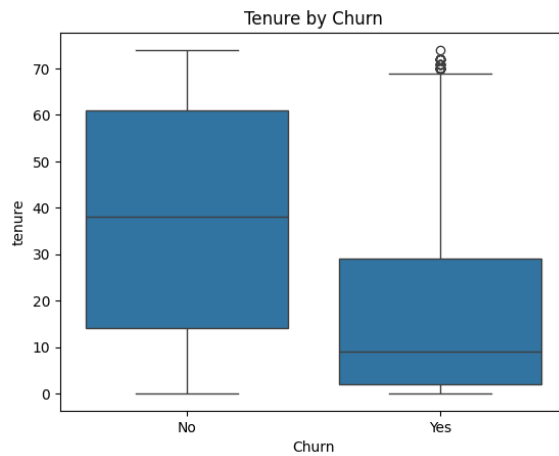
Electronic check users show nearly 48% churn, while automatic payment users churn the least.

Insight: Customers without financial lock-in are more likely to churn.

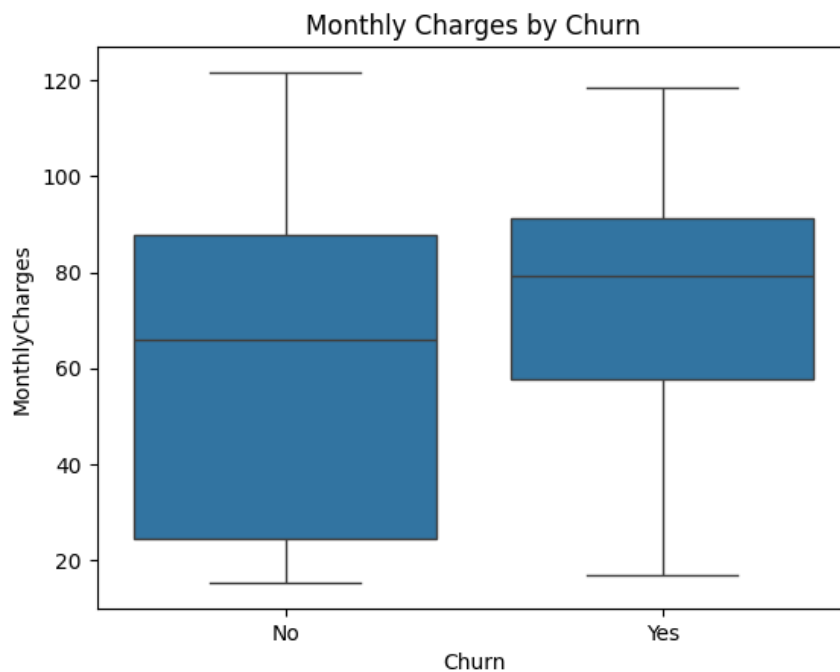


Numeric Variables vs Churn

- Churned customers have much lower tenure



- Churned customers pay higher monthly charges



- Churned customers have lower total lifetime spending

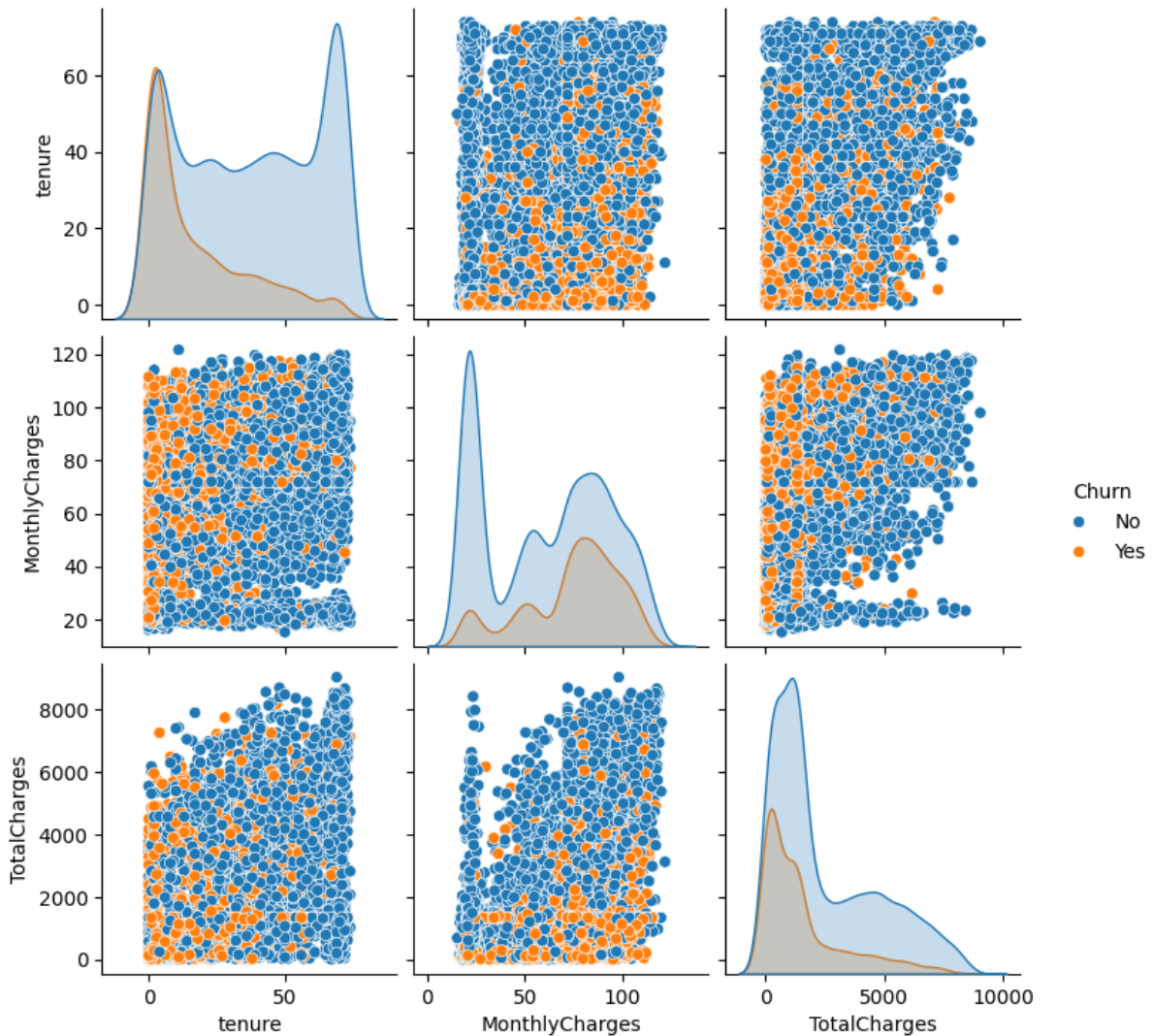
2.5 Multivariate Analysis

Contract + Tenure vs Churn

Month-to-month customers with tenure below 12 months show more than 50% churn. Long-term contracts show very low churn across all tenure groups.

Contract + Payment Method vs Churn

Month-to-month customers using electronic check show the highest churn (around 58%). Long-term contracts show low churn regardless of payment method.



2.6 Key Insights from EDA

- Contract type is the strongest churn driver
- New customers are at highest risk
- Electronic check users are unstable
- Fiber optic service has unusually high churn
- High monthly charges increase churn probability

3. Data Preprocessing

3.1 Methodology

Data preprocessing was performed to ensure clean, reliable input for modeling. Steps included cleaning text, handling missing values, treating anomalies, encoding categorical variables, scaling numeric features, and preparing data for modeling.

3.2 Duplicate Check and Cleaning

- Duplicate records were checked and removed where necessary
- Inconsistent text values were standardized (e.g., Yes/NO/no)
- Currency symbols were removed from monetary columns

3.3 Anomalous Value Treatment

- Records with negative tenure were removed
- Records with negative total charges were removed

3.4 Missing Value Treatment

- MonthlyCharges and TotalCharges had missing values
- Missing values were filled using median imputation

3.5 Outlier Handling

After anomaly removal, no extreme outliers required separate treatment.

3.6 Feature Engineering

- Created tenure groups for EDA purposes
- Converted churn into binary target variable
- One-hot encoded categorical variables

3.7 Class Imbalance Handling

Churn rate is around 28%. Stratified train-test split was used to preserve class distribution.

Feature Engineering

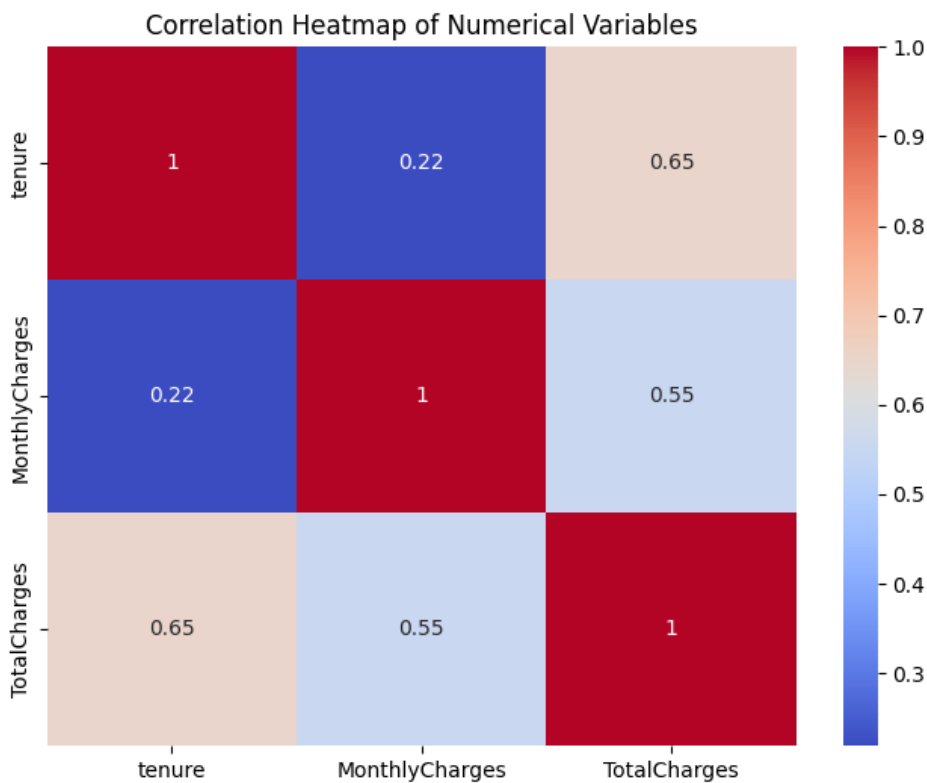
To capture customer lifecycle effects, a new feature was created:

Tenure Group

Tenure was categorized into bands:

- 0–12 months
- 12–24 months
- 24–36 months
- 36–48 months
- 48–60 months
- 60+ months

This feature helped reveal early-stage churn behavior more clearly.



3.8 Feature Scaling

Numeric variables were standardized using StandardScaler for logistic regression.

3.9 Data Leakage Prevention

Scaling and transformations were fitted only on training data and applied to test data.

Key Insights from EDA

- Contract type is the strongest churn driver
- Early-tenure customers are most vulnerable
- Electronic check users are unstable
- Fiber optic service shows unusually high churn
- Higher monthly charges increase churn probability

4. Model Building – Baseline Model

4.1 Methodology and Algorithm Choice

Logistic Regression was selected as the baseline model due to its simplicity, interpretability, and suitability for binary classification problems.

4.2 Metric of Choice and Rationale

Recall was chosen as the primary metric because missing a churner is more costly than incorrectly flagging a non-churner. The business goal is to identify as many potential churners as possible.

4.3 Model Performance

Baseline logistic regression achieved:

- Accuracy $\approx 80\%$
- Precision (churn) $\approx 67\%$
- Recall (churn) $\approx 56\%$
- F1-score $\approx 61\%$
- ROC-AUC ≈ 0.85

Confusion Matrix Summary:

- True Positives: 434
- False Negatives: 339
- False Positives: 218
- True Negatives: 1809

4.4 Model Interpretation

The model correctly classifies about 80% of customers and identifies more than half of actual churners. Even as a simple baseline, the model shows strong discriminatory power and is suitable as an early-warning churn detection system.

5. MODEL BUILDING – ADVANCED MODELS

To improve predictive performance, multiple advanced machine learning models were implemented:

- Decision Tree
- Random Forest
- Bagging Classifier
- AdaBoost
- Gradient Boosting

Each model was evaluated using the same metrics for fair comparison.

Accuracy	Precision	Recall	F1	ROC_AUC	
Decision Tree	0.708214	0.472222	0.483829	0.477955	0.639320
Random Forest	0.796786	0.666124	0.529107	0.589762	0.826689
Bagging	0.771071	0.614187	0.459250	0.525537	0.775884
AdaBoost	0.801429	0.669797	0.553687	0.606232	0.841507
Gradient Boosting	0.816071	0.705414	0.573092	0.632405	0.850218

6. MODEL PERFORMANCE IMPROVEMENT (HYPERPARAMETER TUNING)

6.1 Tuning Strategy

Due to computational constraints, **manual hyperparameter tuning** was adopted instead of exhaustive grid search. Key parameters affecting bias-variance tradeoff were adjusted iteratively.

This approach:

- Reduced overfitting
- Improved generalization

- Ensured reproducibility

6.2 Before vs After Comparison

Accuracy	Precision	Recall	F1	ROC_AUC	
DT - Default	0.708214	0.472222	0.483829	0.477955	0.639320
DT - Tuned	0.794286	0.671902	0.498060	0.572065	0.822097
RF - Default	0.796786	0.666124	0.529107	0.589762	0.826689
RF - Tuned	0.807857	0.692308	0.547219	0.611272	0.845000
Ada - Default	0.801429	0.669797	0.553687	0.606232	0.841507
Ada - Tuned	0.782500	0.700980	0.369987	0.484335	0.834854
GB - Default	0.816071	0.705414	0.573092	0.632405	0.850218
GB - Tuned	0.814286	0.699213	0.574386	0.630682	0.850530

Key observations:

- Decision Tree showed substantial improvement after tuning
- Random Forest achieved consistent gains
- Gradient Boosting remained the most stable and best-performing model

7. MODEL COMPARISON AND FINAL MODEL SELECTION

Among all models tested, **Gradient Boosting Classifier** demonstrated:

- Highest ROC-AUC
- Strong F1-score
- Balanced Precision-Recall tradeoff
- Stable performance before and after tuning

8. Conclusion

This project demonstrates how predictive analytics can transform churn management from a reactive to a proactive strategy. By leveraging customer data and advanced machine learning models, AlphaCom can identify high-risk customers early and deploy targeted retention strategies.

The final Gradient Boosting model provides a robust foundation for churn prediction and can significantly reduce avoidable revenue loss while improving customer lifetime value.

9. Appendix

All raw code, extended tables, and detailed model outputs are provided in the accompanying HTML notebook for verification and reproducibility.

- Detailed model outputs
- Confusion matrices
- Full metric tables
- HTML notebook reference

Reference: [Aryaman_CustomerChurn_FinalReport.html](#)